

Esercizio n°1 (7 punti)

Dato il grafo degli stati in figura, scrivere la tabella delle transizioni degli stati usando FF-JK e le relative espressioni algebriche semplificate con le mappe di Karnaugh.

Q(t)	Q(t+1)	J	K
0	0	0	0
0	1	1	0
1	0	0	1
1	1	0	0

STATO	STATO SUCCESSIVO	
	X=0	X=1
S0	S1/0	S0/0
S1	S1/0	S2/0
S2	S0/1	S1/0

A	B	X	A'	JA	KA	B'	JB	KB	Z
0	0	0	0	0	0	1	1	0	0
0	0	1	0	0	0	0	0	0	0
0	1	0	0	0	0	1	0	0	0
0	1	1	1	1	0	0	0	1	0
1	0	0	0	0	1	1	0	1	1
1	0	1	0	0	1	0	0	0	0
1	1	0	0	0	0	0	0	0	0
1	1	1	0	0	0	0	0	0	0

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Esercizio n°2 (6 punti)

Si consideri una gerarchia di memoria a due livelli cache-primaria. L'unità di memoria primaria dispone di 32 B, e la cache contiene 16 parole. E' possibile accedere al singolo byte e la memoria è suddivisa in blocchi da 2 B. Indicare, specificando l'ampiezza e la funzione dei diversi campi, come vengono interpretati gli indirizzi di memoria primaria secondo il metodo associativo su insiemi a due vie.

Metodo associativo a 2 vie

MP = MEMORIA PRIMARIA 32 B = 2⁵ B = 5 bit

MC = MEMORIA CACHE 16 PAROLE = 2⁴ = 4 bit

dim MB = DIMENSIONE BLOCCHI 2 B = 1 bit

2 VIE

Block Frame

TAG 2 bit	SET INDEX 2 bit	OFFSET 1 bit
BF = TAG + SI + OF = 5 bit		
SI = MC / dim MB = 2 VIE		
2 ⁴ / 2 = 2 ² = 4		

Esercizio n°3 (8 punti)

Scrivere un frammento di codice MIPS che moltiplichi per 2N un valore presente nel registro \$4. Si consideri N intero non negativo

```
lw $5, $0      $5 <- N
move $8, $0    $8 <- i = 0
loop: beq $8, $5, exit if i == N -> exit
mul $7, $7, 2  $7 <- 2 * moltiplicato N volte
addi $8, $8, 1
i FOR
exit: mul $7, $7, $4  k = 2^N
...
```

Esercizio n°4 (6 punti)

Scrivere una funzione Assembly MIPS retta che calcoli, memorizzandoli nelle aree di memoria dette simbolicamente m e q, i parametri di equazione y = m * x + q, date le coordinate di due punti sul piano cartesiano (x0, y0) e (x1, y1) nei registri \$4-\$5 e \$6-\$7. Si consideri che i due punti non possono giacere su rette coincidenti o parallele all'asse x o all'asse y. Si faccia uso della funzione div che riceve in \$4 e \$5 il dividendo e divisore e metta in \$6 il quoziente. Si ricordano le formule:

$$m = \frac{(y_1 - y_0)}{(x_1 - x_0)}$$
$$q = \frac{x_1 \cdot y_0 - x_0 \cdot y_1}{x_1 - x_0}$$

addi \$29, \$29, -20 # salvataggio contesto

sw \$9, 0(\$29)

sw \$10, 4(\$29)

sw \$11, 8(\$29)

sw \$8, 12(\$29)

sw \$31, 16(\$29)

move \$9, \$4 # copia di x0

move \$10, \$5 # copia di y0

move \$11, \$6 # copia di x1

sub \$4, \$7, \$10 # preparazione dividendo (y1 - y0)

sub \$5, \$11, \$9 # preparazione divisore (x1 - x0)

jal div

sw \$6, m(\$0)

mul \$4, \$11, \$10 # preparazione dividendo (x1 * y0)

mul \$8, \$9, \$7 # preparazione dividendo (x0 * y1)

sub \$4, \$4, \$8 # preparazione dividendo (x1 * y0 - x0 * y1)

jal div

sw \$6, q(\$0)

exit: move \$4, \$9 # ripristino x0

move \$5, \$10 # ripristino y0

move \$6, \$11 # ripristino x1

lw \$9, 0(\$29)

lw \$10, 4(\$29)

lw \$11, 8(\$29)

lw \$8, 12(\$29)

lw \$31, 16(\$29)

addi \$29, \$29, 20 # ripristino del contesto

jr \$31 # ritorno al chiamante

SUB \$4, \$7, \$5

SUB \$5, \$6, \$4

JAL div

MOVE \$15, \$6

MOVE \$11, \$4

MOVE \$12, \$5

MOVE \$13, \$6

MOVE \$14, \$7

SUB \$4, \$14, \$12

SUB \$5, \$13, \$11

JAL div

MUL \$12, \$13, \$12 x1 * y0

MUL \$14, \$14, \$11 x0 * y1

SUB \$4, \$7, \$14

SUB \$5, \$6, \$13

JAL div

J \$31

ESERCIZIO n°5 (3 PUNTI)

Sia dato il formato IEEE 754 su valori a 32 bit. Rappresentare il valore 162.25.

S	ESPONENTE 8 BIT	FRAZIONARIA 23 BIT
0	10000110	010001001000...0

$(162,25)_{10} \rightarrow (10100010,01)_2 \rightarrow 1,010001001 \cdot 2^7$ CON ECCESSO 127

ESPONENTE = 7 + ECCESSO = 7 + 127 = (134)₁₀ = (10000110)₂

$162 : 2 = 81$
 $81 : 2 = 40$
 $40 : 2 = 20$
 $20 : 2 = 10$
 $10 : 2 = 5$
 $5 : 2 = 2$
 $2 : 2 = 1$
 $1 : 2 = 0$

$0,25 : 2 = 0,125$
 $0,125 : 2 = 0,0625$
 $0,0625 : 2 = 0,03125$
 $0,03125 : 2 = 0,015625$
 $0,015625 : 2 = 0,0078125$
 $0,0078125 : 2 = 0,00390625$
 $0,00390625 : 2 = 0,001953125$
 $0,001953125 : 2 = 0,0009765625$
 $0,0009765625 : 2 = 0,00048828125$
 $0,00048828125 : 2 = 0,000244140625$
 $0,000244140625 : 2 = 0,0001220703125$
 $0,0001220703125 : 2 = 0,00006103515625$
 $0,00006103515625 : 2 = 0,000030517578125$
 $0,000030517578125 : 2 = 0,0000152587890625$
 $0,0000152587890625 : 2 = 0,00000762939453125$
 $0,00000762939453125 : 2 = 0,000003814697265625$
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